CUSTOMER SEGMENTATION AND PURCHASING BEHAVIOUR ANALYSIS

Using Unsupervised Machine Learning

Dataset: UCI Online Retail

Methods: K-Means, Agglomerative Clustering and DBSCAN

Analytical report documenting data preparation, customer-level feature engineering, dimensionality reduction, clustering model comparison, final segment selection and business interpretation.

Prepared from the completed project analysis and observed model outputs.

Executive Summary

This project applies unsupervised machine learning to the UCI Online Retail dataset to identify groups of customers with similar purchasing behaviour. The original dataset contains 541,909 transaction records from a UK-based online retailer covering December 2010 to December 2011.

After removing duplicate, cancelled and invalid transactions and retaining transactions linked to identifiable customers, 392,692 valid transactions remained. These transactions represented 4,338 customers. Six customer-level behavioural features were constructed: Recency, Frequency, Monetary, Total Quantity, Product Diversity and Average Order Value.

Because the customer features were strongly right-skewed, a log transformation was applied before standardisation. PCA was then used to visualise the customer space, while clustering was performed using the complete six-dimensional standardised feature set.

K-Means, Agglomerative Clustering and DBSCAN were compared. K-Means and Agglomerative Clustering both selected two clusters during model-selection testing. DBSCAN achieved stronger Silhouette and Davies-Bouldin scores at the selected setting, but its solution was dominated by one cluster containing 4,257 customers and a second cluster containing only three customers, with 78 observations treated as noise.

K-Means with two clusters was therefore selected as the final model because it provided the strongest overall balance between internal validation, balanced segment sizes and business interpretability.

Final segmentation: Cluster 0 contained 1,924 customers and showed substantially higher purchasing activity and monetary contribution. Cluster 1 contained 2,414 customers and showed lower frequency, lower monetary contribution and longer time since the last purchase.

1. Project Objective

The purpose of the analysis was to use unsupervised learning techniques to discover hidden structure within customer purchasing behaviour. Rather than predicting a predefined target, the analysis seeks to identify naturally occurring groups that can support differentiated retail decision-making.

The specific objectives were to:

• assess the quality and structure of the transaction data;
• create meaningful customer-level behavioural features;
• compare multiple clustering approaches;
• evaluate the resulting clusters using internal validation measures;
• select a practically useful final segmentation; and
• interpret the resulting customer groups for business use.

2. Dataset and Data Description

The project uses the UCI Online Retail dataset (Dataset ID 352). The dataset contains transaction records from a UK-based online retailer. The original data contained 541,909 records and eight variables.

Variable	Description
InvoiceNo	Invoice identifier
StockCode	Product identifier
Description	Product description
Quantity	Quantity purchased
InvoiceDate	Transaction date and time
UnitPrice	Price per item
CustomerID	Customer identifier
Country	Customer country

Source: UCI Machine Learning Repository, Online Retail dataset. The project used the supplied retail.xlsx file.

3. Data Quality Assessment

The initial inspection identified missing customer identifiers, duplicate transactions, cancellation invoices, negative quantities and non-positive prices. These issues were considered in relation to the intended customer-level clustering task.

Issue	Observed count
Missing CustomerID	135,080
Duplicate rows	5,268
Cancellation invoices	9,288
Negative quantities	10,624
Zero/negative prices	2,517
Missing descriptions	1,454

Cleaning decisions

Exact duplicates were removed. Records without CustomerID were excluded because a customer identifier is required for customer-level segmentation. Cancellation invoices were removed, as were transactions with non-positive quantities or prices. Missing descriptions were not used as a deletion criterion because product descriptions were not required for the selected customer-level feature set.

After cleaning, 392,692 transaction records remained. The cleaned data covered the period from 1 December 2010 to 9 December 2011 and contained 4,338 unique customers. The resulting transaction value was 8,887,208.89.

4. Customer-Level Feature Engineering

The transaction data was transformed into one row per customer. The feature design combines RFM-style measures with additional indicators of purchase volume and product breadth.

Feature	Definition / interpretation
Recency	Days since the customer's most recent purchase
Frequency	Number of distinct invoices
Monetary	Total transaction value
TotalQuantity	Total quantity purchased
ProductDiversity	Number of distinct stock codes purchased
AverageOrderValue	Average value of the customer's orders

The customer feature dataset contained 4,338 customers and six clustering variables. The distributions were strongly skewed, with very large values visible for monetary value, quantity and order value. This was addressed before clustering.

5. Preprocessing

A log1p transformation was applied to each clustering feature to reduce the influence of extreme values. The transformed variables were then standardised using StandardScaler so that the clustering algorithms operated on comparable feature scales.

The first row of the standardised feature matrix and the resulting means/standard deviations confirmed that the preprocessing pipeline was operating as intended. No target variable was introduced; the analysis remained fully unsupervised.

6. Principal Component Analysis

PCA was used to create a two-dimensional representation of the six-dimensional customer feature space. The clustering models themselves were not fitted to the two PCA components; the full six-feature standardised matrix was retained for clustering.

PC1 explained 65.76% of variance and PC2 explained 17.39%, giving 83.15% cumulative variance for the two-dimensional projection.

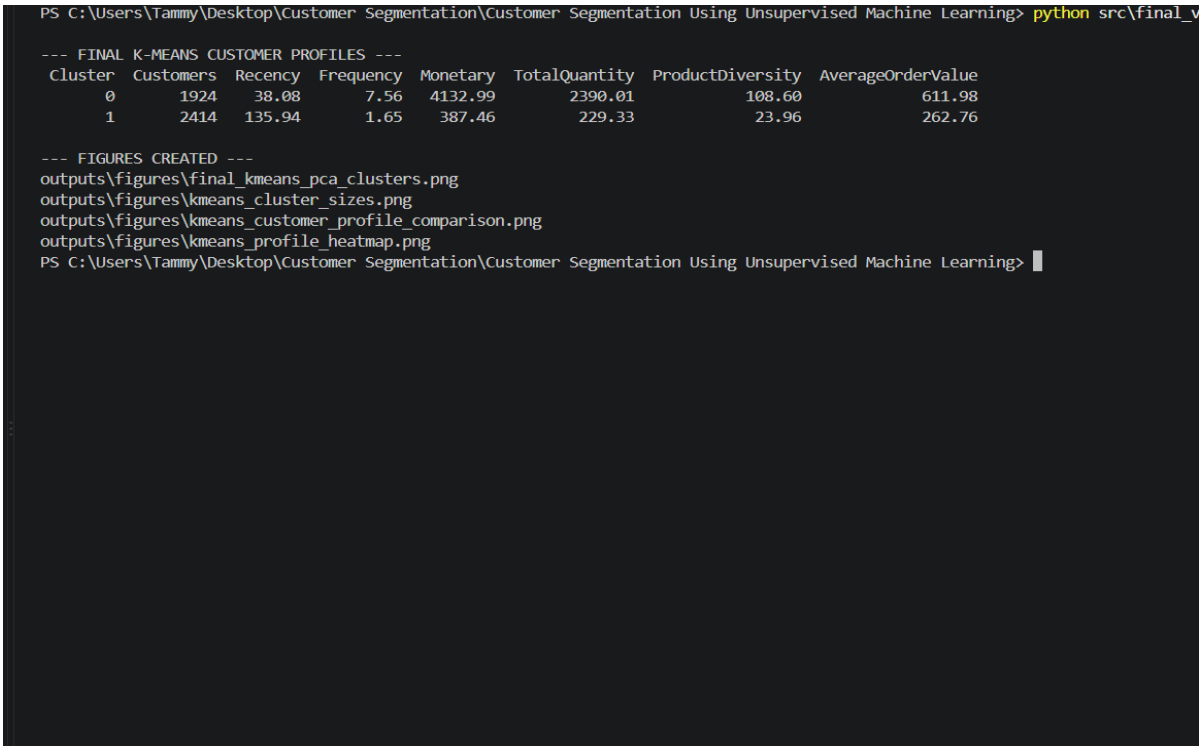


Figure 1. Final K-Means customer segmentation projected onto the first two principal components.

The projection shows a visible separation between the two final groups, particularly along PC1, although observations overlap around the boundary. This supports the use of quantitative clustering validation rather than relying on visual separation alone.

7. K-Means Model Selection

K-Means was evaluated for K values from 2 to 10. Three internal measures were used: Silhouette Score, Davies-Bouldin Index and Calinski-Harabasz Index.

K	Silhouette	Davies-Bouldin	Calinski-Harabasz
2	0.3563	1.0550	3183.4055
3	0.2609	1.2331	2554.2734
4	0.2261	1.3313	2151.5124
5	0.2270	1.2903	1949.8504
6	0.2124	1.2987	1795.4173
7	0.2064	1.2959	1648.8139
8	0.2060	1.3168	1545.8596
9	0.2037	1.3030	1459.8290
10	0.1954	1.3112	1392.2332

All three measures selected K=2 within the tested range. The final K-Means configuration used n_clusters=2, random_state=42 and n_init=20.

8. Agglomerative Clustering

Ward-linkage Agglomerative Clustering was evaluated from two through ten clusters. The strongest configuration was again K=2, with a Silhouette Score of 0.3063, Davies-Bouldin Index of 1.1336 and Calinski-Harabasz Index of 2563.2299.

9. DBSCAN

DBSCAN was investigated using a five-nearest-neighbour distance curve and a range of epsilon values. The selected configuration was eps=1.1 with min_samples=5.

At eps=1.1, DBSCAN identified two clusters and labelled 78 observations as noise. Its Silhouette Score was 0.4724 and Davies-Bouldin Index was 0.4735. However, the cluster profile revealed that 4,257 customers belonged to one cluster and only three customers belonged to the second cluster.

This is an important example of why internal clustering metrics should not be interpreted in isolation. A model can produce strong separation scores while still generating a segmentation that is too imbalanced to be useful for the intended business purpose.

10. Final Model Comparison and Selection

Model	Clusters	Noise	Silhouette	DB Index	CH Index
K-Means	2	0	0.3563	1.0550	3183.4055
Agglomerative	2	0	0.3063	1.1336	2563.2299
DBSCAN	2	78	0.4724	0.4735	19.3966

Final selection: **K-Means with two clusters.**

K-Means was selected because it offered the strongest overall balance between internal validation, balanced segment sizes and business interpretability. Agglomerative clustering was weaker across the three validation measures. DBSCAN produced higher Silhouette and lower Davies-Bouldin values, but its solution was dominated by one very large cluster and a three-customer cluster, limiting its analytical usefulness.

Comparability caveat: DBSCAN metrics were calculated after excluding its 78 noise observations, whereas K-Means and Agglomerative metrics used all 4,338 customers. Therefore, the DBSCAN scores are not perfectly apples-to-apples with the other two models.

11. Final Customer Profiles

Measure	Cluster 0	Cluster 1
Customers	1,924	2,414
Recency	38.08	135.94
Frequency	7.56	1.65
Monetary	4,132.99	387.46
Total Quantity	2,390.01	229.33
Product Diversity	108.60	23.96
Average Order Value	611.98	262.76

Cluster 0 — High-Value Active Customers

Cluster 0 contains 1,924 customers. The group has a mean recency of 38.08 days, mean frequency of 7.56 orders and mean monetary value of 4,132.99. It also has higher total quantity, product diversity and average order value. The profile therefore represents the stronger historical purchasing group in the dataset.

Cluster 1 — Lower-Value Less-Active Customers

Cluster 1 contains 2,414 customers. The group has a mean recency of 135.94 days, frequency of 1.65 orders and monetary value of 387.46. Its lower product diversity, quantity and average order value reinforce the interpretation of a less-active purchasing profile.

The labels describe observed behaviour in the analysed period. They should not be interpreted as permanent customer identities or as evidence of future behaviour.

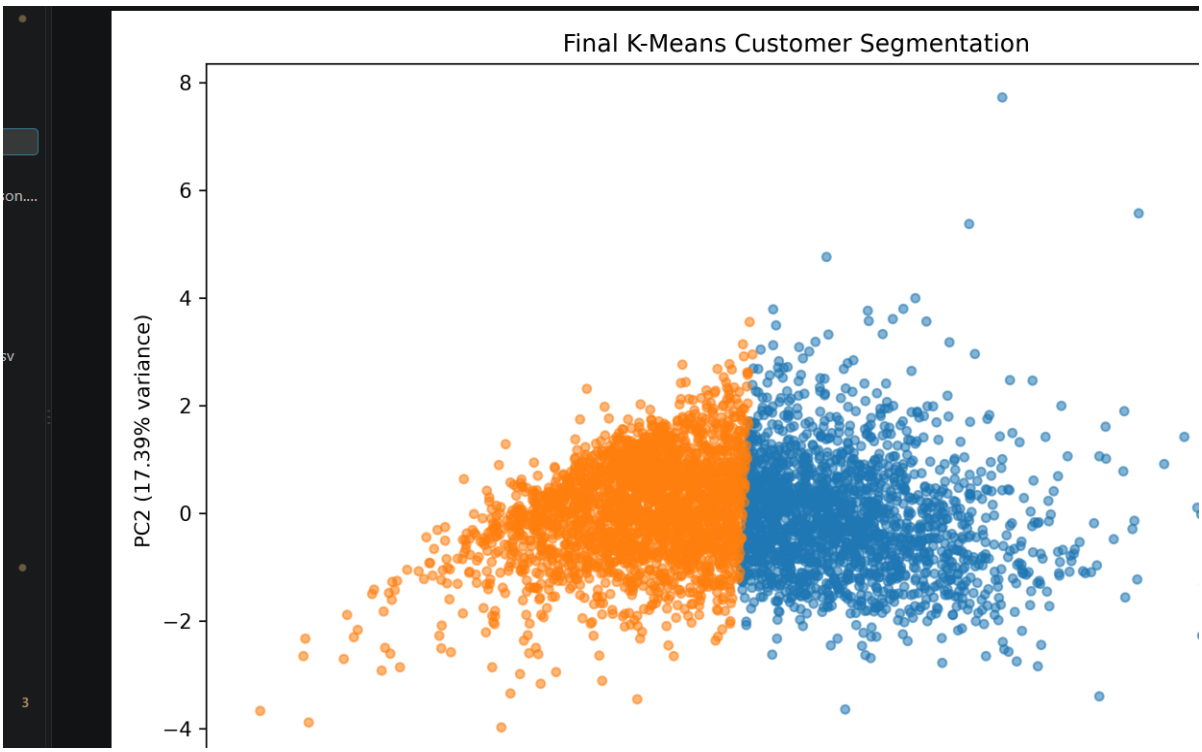


Figure 2. Comparison of mean customer-level behavioural measures across the final K-Means clusters.

12. Segment Visualisation

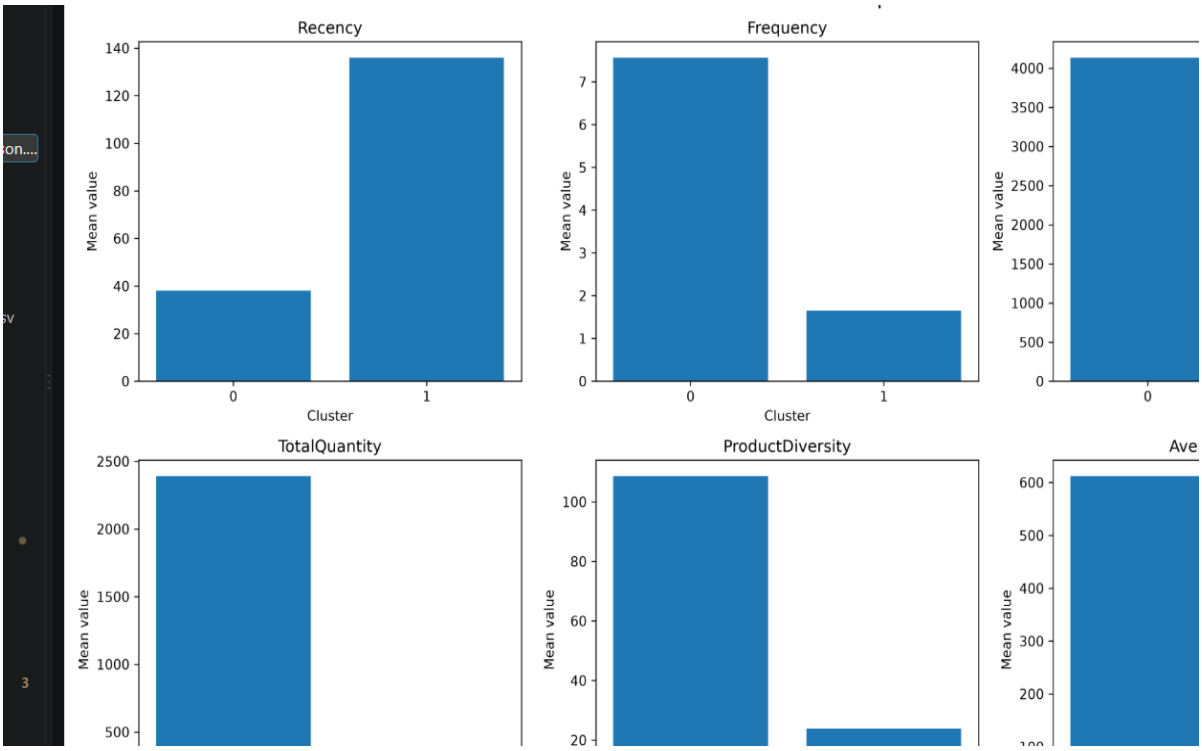


Figure 3. Relative customer profiles using standardised cluster means.

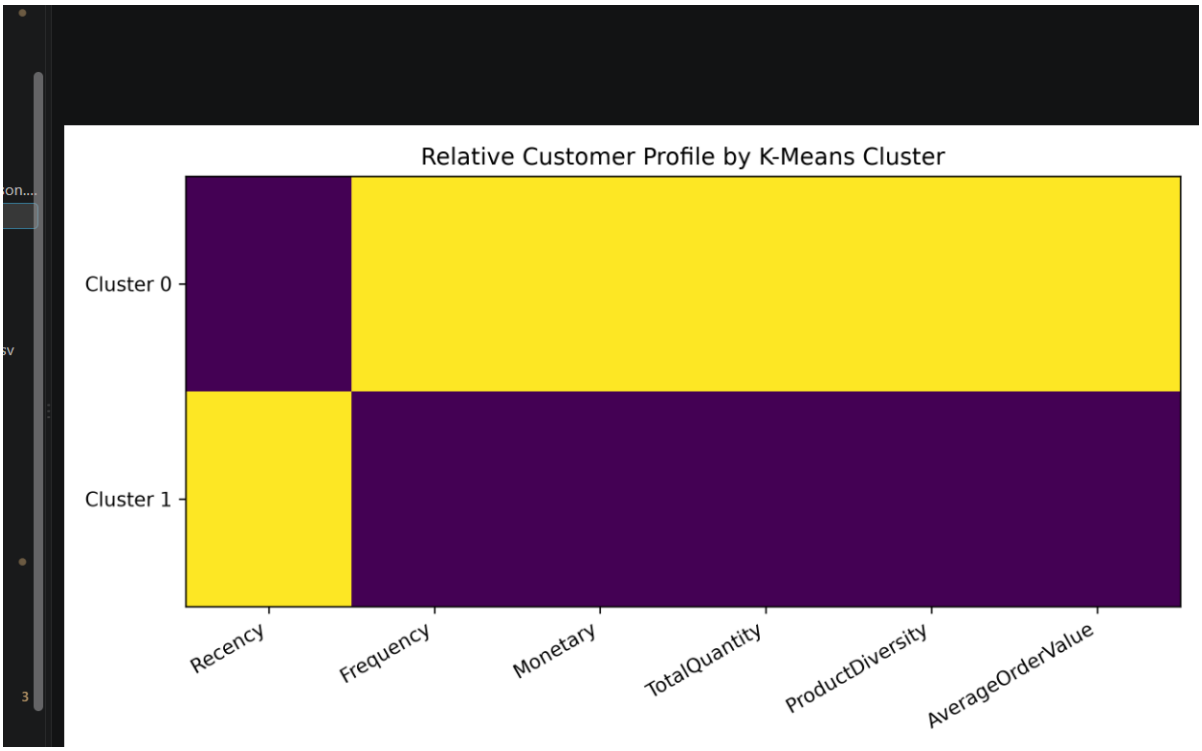


Figure 4. Number of customers assigned to each final K-Means cluster.

13. Business Implications

The segmentation indicates that the customer base contains materially different purchasing profiles. Treating all customers as a single population would therefore hide meaningful differences in recency, purchasing frequency, spending and product breadth.

For the High-Value Active segment: the business can prioritise retention, relationship building and relevant cross-selling opportunities. The segment's stronger historical activity makes it important to protect from disengagement.

For the Lower-Value Less-Active segment: the business can investigate approaches designed to encourage repeat purchasing and engagement. Further analysis would be required before selecting specific campaigns or offers.

The recommended use of these segments is as a decision-support layer. Segment membership should be refreshed periodically because customer behaviour can change over time.

14. Limitations

The analysis reflects historical transactions within the available observation period and does not capture all dimensions of customer value. The resulting clusters depend on the selected features, aggregation definitions, log transformation and standardisation. PCA provides a visual projection rather than a complete representation of the six-dimensional clustering space. In addition, the DBSCAN validation scores exclude noise observations, which limits direct comparison with the other models.

The selected feature set does not include demographic information, campaign response, profitability, customer-service interactions or other contextual variables. Consequently, the segments should be interpreted specifically as purchasing-behaviour segments.

15. Future Work and Action Plan

The segmentation can be extended from an exploratory analysis into a repeatable customer analytics process through the following actions:

1. Refresh the segmentation: rerun the analysis periodically as new transactions arrive to identify changes in customer behaviour.

2. Expand behavioural features: test purchase intervals, category-level spending, return behaviour and other measures that may add business value.

3. Test stability: compare cluster assignments across different time windows and samples to determine how stable the segmentation is.

4. Connect to outcomes: where labelled business data become available, test whether the segments differ in campaign response, retention or other measurable outcomes.

5. Evaluate interventions: compare segment-specific actions using measurable business KPIs rather than assuming that a cluster automatically represents a successful marketing strategy.

16. Reproducibility and Project Artefacts

The project includes standalone Python scripts under **src/**, generated tables under **outputs/tables/**, visualisations under **outputs/figures/**, and an end-to-end Jupyter notebook under **notebooks/customer_segmentation_analysis.ipynb**. The README documents the project structure, workflow and main results.

The analytical workflow is reproducible from the project root using the scripts documented in the README. The final report is based on the observed outputs generated during the analysis.

17. Conclusion

This project demonstrates an end-to-end unsupervised learning workflow for customer segmentation using real-world retail transaction data. After data cleaning and customer-level feature engineering, K-Means, Agglomerative Clustering and DBSCAN were evaluated using multiple internal validation measures.

K-Means with two clusters was selected because it produced a balanced and interpretable segmentation. The final groups differ clearly in purchasing recency, frequency, monetary value, quantity, product diversity and average order value. These differences provide a practical basis for differentiated customer analysis while leaving room for future validation against business outcomes.

> **Final analytical outcome:** the project identifies two actionable purchasing-behaviour profiles from the 4,338-customer modelling population and demonstrates why model selection should combine statistical validation with practical interpretability.

Appendix A — Main Analytical Outputs

The following artefacts were generated during the analysis and are used as supporting evidence for the report:

Artefact	Purpose
customer_features.csv	Customer-level behavioural features
clustering_features.csv	Log-transformed and standardised clustering matrix
kmeans_selection.csv	K-Means K-selection results
agglomerative_selection.csv	Agglomerative K-selection results
dbscan_evaluation.csv	DBSCAN parameter evaluation
final_model_comparison.csv	Final model comparison
kmeans_customer_clusters.csv	Final K-Means customer assignments
final_kmeans_pca_clusters.png	Final PCA cluster visualisation
kmeans_cluster_sizes.png	Final cluster-size visualisation
kmeans_customer_profile_comparison.png	Final behavioural profile comparison
kmeans_profile_heatmap.png	Relative standardised profile visualisation

Recommended repository documentation: README.md and the complete Jupyter notebook customer_segmentation_analysis.ipynb.